Synoptic CB: Manual Redox Probes

2025 Samples

2026-04-14

Measurements taken with SWAP Redox Probes & Calomel Ref probe

Protocol: https://docs.google.com/document/d/

1dL2ZPhLFBAZJmVBCfKxup_KOYY96qA5IWfJmQen50pY/edit

Units = mV

##Add Required Packages

##Setup

```
###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
dat <- read.csv("Raw Data/COMPASS_Synoptic_CB_Redox_2025_Raw.csv")
soildat <- read.csv("Raw Data/CB_Soil_pH.csv")

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_Synoptic_CB_Redox_2025_processed.csv"

# ##Fix Sample Zone and Site naming
# soildat <- soildat %>%
#   mutate(Zone = case_when(Site == "SWH" & Zone == "Upland" ~ "Swamp",
#                           Site == "SWH" & Zone == "Upland-Control" ~ "Upland",
#                           TRUE ~ Zone),
#          Site = ifelse(Site == "Gcrew", "GCW", Site))

##Fix Depth mislabelled as 45, should be 35
dat <- dat %>%
  mutate(Depth = ifelse(Depth == 45, 35, Depth))

##Remove space from GWI site name
dat$Site <- gsub(" ", "", dat$Site) ## Remove space in GWI site name

##For November, Depths were labelled 10, 25, 30 but should be 10, 20, 35
# dat <- dat %>%
#   mutate(Depth = case_when(Month == "November" & Depth == 25 ~ 20,
```

```
#                             Month == "November" & Depth == 30 ~ 35,
#                            TRUE ~ Depth))

##Add column for SWH topography
dat <- dat %>%
  mutate(Topography = case_when(Site == "SWH" & Zone == "Transition-Hummock" ~ "Hummock",
                                Site == "SWH" & Zone == "Transition-Hollow" ~ "Hollow",
                                TRUE ~ NA),
         Zone = case_when(Site == "SWH" & Zone == "Transition-Hummock" ~ "Transition",
                          Site == "SWH" & Zone == "Transition-Hollow" ~ "Transition",
                          TRUE ~ Zone))

##Std error function
std.error <- function (x, na.rm=FALSE){
  sqrt(var(x, na.rm = na.rm) / sum(!is.na(x)))}
```

Adjust Raw Readings by pH

Need to adjust for pH:

redox = redoxi + [(ph - 5)59 + 224] This is the equation from Megonigal,

Patrick, & Faulkner 1993

They used a calomel electrode and used 224 to adjust to H2,

we use a KCl ref probe, so we need to add 202 instead

```
##Remove unneccesary column in redox data
prelong <- dat %>%
  select (-Campaign)

#switch redox data to long format
datlong <-  as.data.frame(melt(setDT(prelong), id.vars = c("Date", "Month", "Year", "Site", "Zone", "De
                               variable.name = "RP_mV"))
# ##Add in soil pH data
# all_corr <- merge(datlong, soildat, by = c("Site", "Zone"), all = TRUE)
#
# ##Adjust redox for pH
# all_corr <- all_corr %>%
#   mutate(value_corr = value + ((pH - 5)*59 + 202))

##Average across all 5-6 measurements for each sampling Site, Zone, Depth, Month, Year
red1 <- datlong %>%
  group_by(Date, Month, Year, Site, Zone, Depth, Time_24hr, Time_Zone, Topography, Notes) %>%
  summarise(Date = mean(Date),
```

```
            Avg_mV = mean(value, na.rm=TRUE),
            Std_dv = sd(value, na.rm=TRUE),
            Std_err = std.error(value, na.rm=TRUE),
            Notes = first(Notes)
            )

# head(red1)
```

##Write Out Adjusted & Avg'd Data

Vizualize Data